

Last time info. updated: 03-08-2026, where $p_{\text{beat}} = 0.5000$, and $p_{\text{non_arrival}} = 0.5000$

Number of possible states: 20

Possible states generated OK.-

Number of reachable states: 8. Hence, 12 states were discarded.

Reachable states are:

0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

-----P1 RUN OK-----

Transition matrix (P), with arrival probs. = Markov matrix generated OK.-

Row sums (should all be 1, otherwise check that row's probs):

1	1	1	1	1	1	1	1
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-----P2 RUN OK-----

Eigenvalues:

0.0000 + 0.0000i
-0.0000 - 0.0000i
-0.0000 + 0.0000i
1.0000 + 0.0000i
0.0000 + 0.0000i
0.0000 + 0.0000i
-0.0000 + 0.0000i
0.0000 + 0.0000i

Number of eigenvalues equal to 1: 1

Selected eigenvector column: 4

Double checking ($v * P = v$) Difference (should be close to 0): 0.0000000000

Used tolerance: 1.0000e-05

-> v IS the stationary distribution

Theoretical stationary distribution:

0.1250	0.1250	0.1250	0.1250	0.1250	0.1250	0.1250	0.1250
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-----P3 RUN OK-----

For $h=3$; capacity=3:

____SIMULATION 1____

Starting from State ($p_0(1)$):

Set starting state: 1

Visit counts set to 0.-

RESULTS - From State 1 [0 0 0] (empty):

After 1000000 beats:

-e190:

Arrivals: 499219

Effectively seated: 499219
Rejected: 0

Per beat:

Arrivals 0.4992
Effectively seated 0.4992
Rejected: 0.0000

Rejection rate total: 0.0000%
Rejection rate e190: 0.0000%

SIMULATION 2

Starting from State (p0(8)):
Set starting state: 8

Visit counts set to 0.-

RESULTS - From State 8 [1 1 1] (full):
After 1000000 beats:

:
Arrivals: 501162
Effectively seated: 501162
Rejected: 0
-Total:
Arrivals: 501162
Serviced: 501162
Rejected: 0

Per beat:

Arrivals 0.5012
Effectively seated 0.5012
Rejected: 0.0000

-Total:
Arrivals 0.5012
Effectively seated 0.5012
Rejected: 0.0000

Rejection rate total: 0.0000%
Rejection rate: 0.0000%

RSS DIFF. VALUES FOR MY REPORT

Step	Empty start	Full start
10	7.500000e-02	9.500000e-02
1000	2.220000e-04	1.760000e-04
100000	9.791600e-06	6.341400e-06
1000000	1.521378e-06	2.845860e-06

RANKING OF STATES

Rank	State	Theory	Empty start	Full start	Cum. theory	Cum. empty	Cum. full
1	110	12.5000	12.5112	12.5437	12.5000	12.5112	12.5437
2	111	12.5000	12.4040	12.5298	25.0000	24.9152	25.0735

3	001	12.5000	12.5425	12.4746	37.5000	37.4577	37.5481
4	011	12.5000	12.5112	12.5437	50.0000	49.9689	50.0918
5	010	12.5000	12.4955	12.4990	62.5000	62.4644	62.5908
6	000	12.5000	12.5290	12.3665	75.0000	74.9934	74.9573
7	100	12.5000	12.5424	12.4746	87.5000	87.5358	87.4319
8	101	12.5000	12.4642	12.5681	100.0000	100.0000	100.0000

-----P4 RUN OK-----

Theoretical long-run occupancy shares (from v):

P(all bays empty) = 0.1250
 P(1 bay busy) = 0.3750
 P(2 bays busy) = 0.3750
 P(full) = 0.1250
 Check (should sum to 1): 1.0000
 Mean number of busy bays = 1.5000 of 3
 Long-run utilisation = 50.00%

The expected parameters in the theoretical long run p/plane-p/beat are as follows:

Note: arrival rates were estimated from ADL Airport data using Python.

Aircraft	Arrivals	Rejected	Serviced
	0.5000	0.0000	0.5000
Total	0.5000	0.0000	0.5000

The expected parameters in the theoretical long run p/plane-p/day are as follows:

Aircraft	Arrivals	Rejected	Serviced
E190	72.0000	0.0000	72.0000

-----P5 RUN OK-----

Plot generated OK.-

-----P6 RUN OK-----

-----P9 RUN OK-----

Correctly saved

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